Fundamentals of Robotics and Applications

Unit 5: Trajectory Planning and Robot Applications

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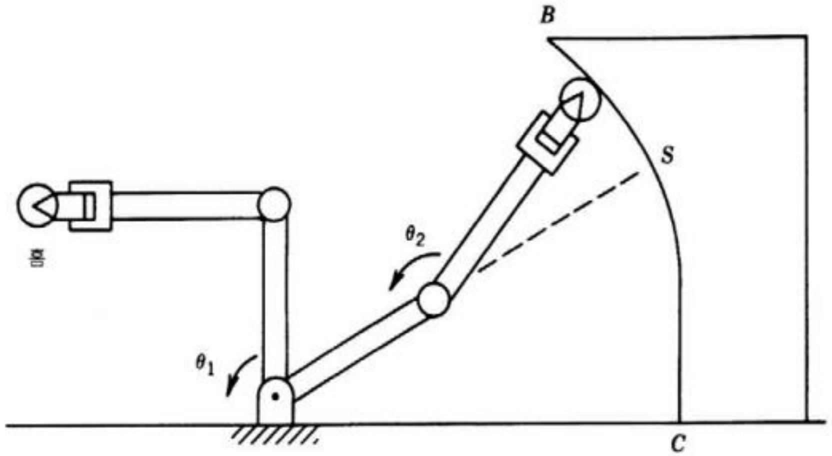
Trajectory planning

- Path and trajectory planning means the way that a robot is moved from one location to another in a controlled manner.
- The trajectory or path refers to a time history of position, velocity and acceleration of all possible movements of the manipulator.
- The sequence of movements for a controlled movement between motion segment, in straight line or in sequential motions.
- It requires the use of both kinematics and dynamics of robots.
- Goal is to generate the reference inputs to the motion control system which ensures that the manipulator executes the planned trajectory.



As shown in Fig. 5.A1, the basic problem is to move the manipulator from an initial position to some desired final position.

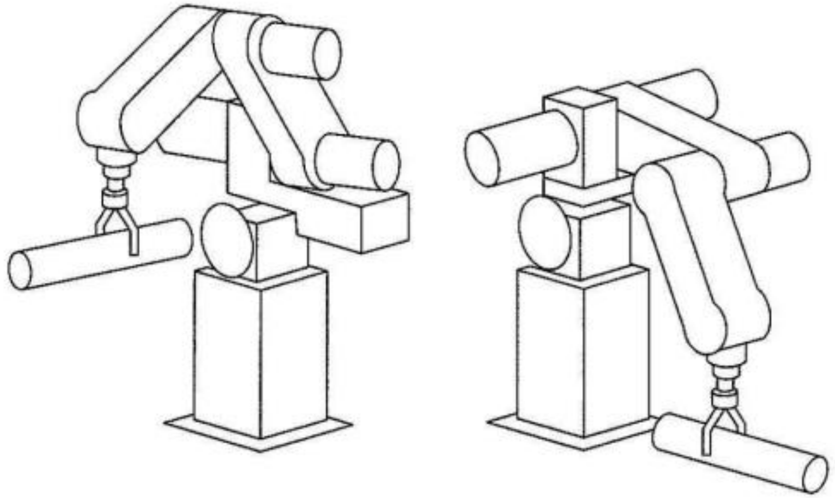


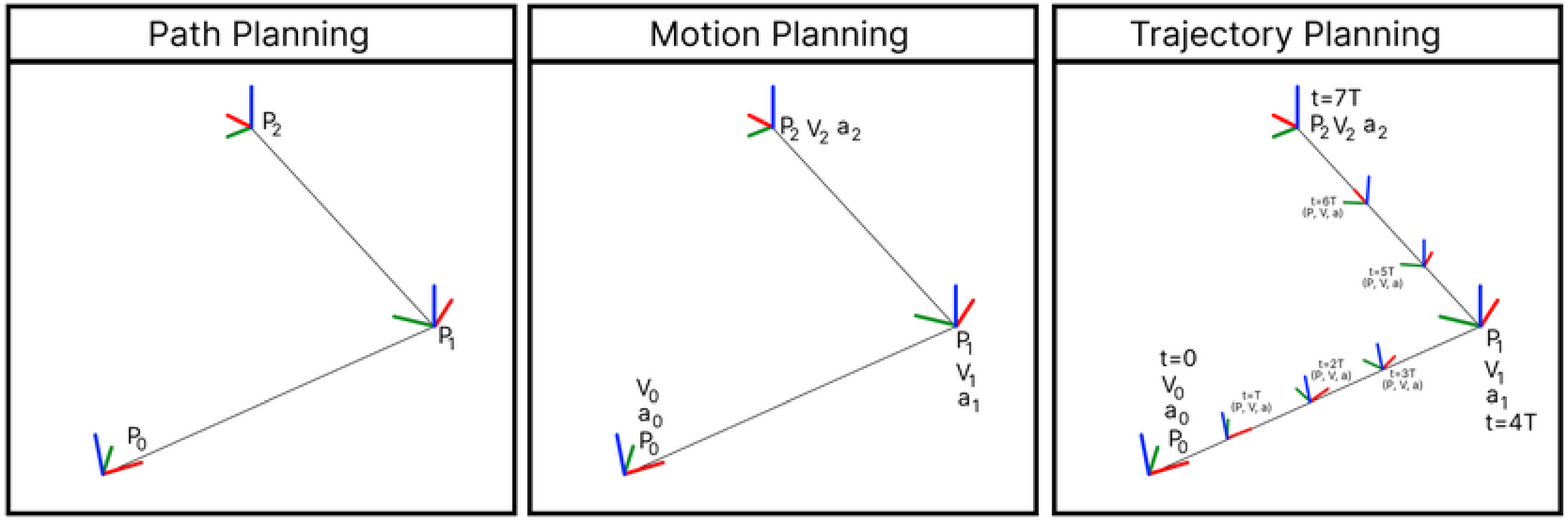
FIGURE 5.A1 : In executing a trajectory, a manipulator moves from its initial position to a desired goal position in a smooth manner.

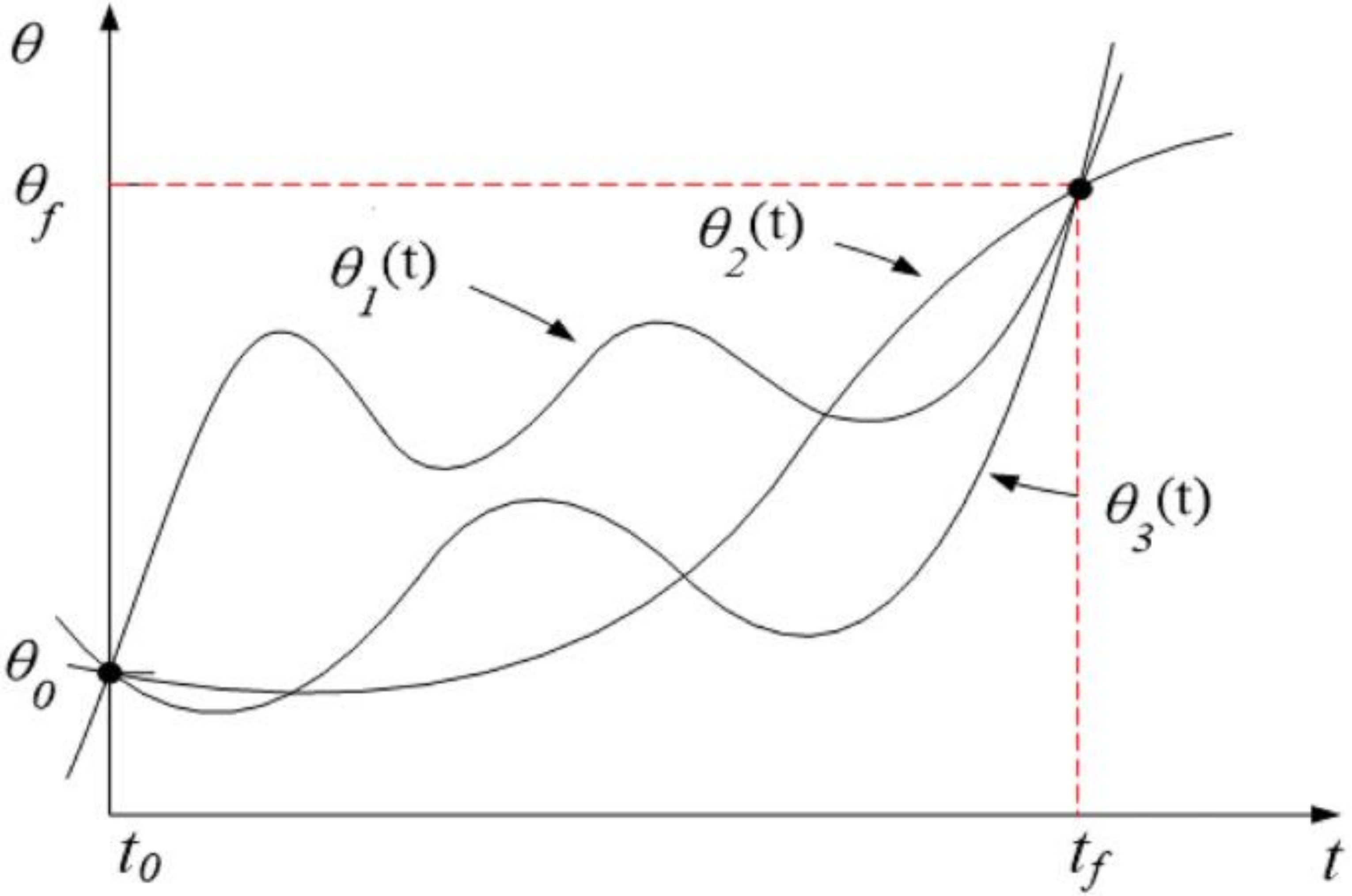
In general, this motion involves both a change in orientation and a change in position of the tool relative to the station.

Trajectory planning

Path Vs Trajectory

- Path: A path is the locus of points to be traversed by manipulator/robot to execute specified task. A path is purely geometric description of motion, without regard to the timing.
- Trajectory: A trajectory is a path with specified qualities of motion, i.e., a path on which a time law is specified in terms of velocity and/or accelerations at each point.





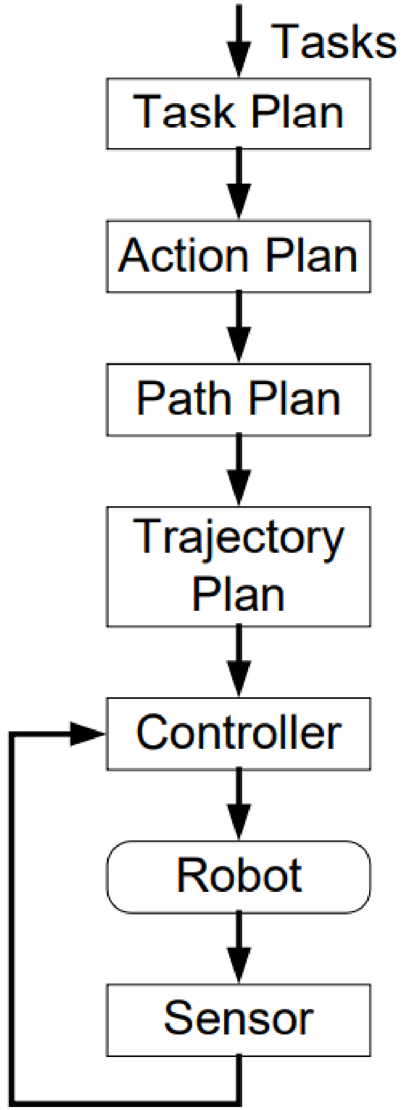
θ
θ_f
$\theta_1(\mathrm{t})$
$\theta_2(\mathrm{t})$
$\theta_3(\mathrm{t})$
θ_0
t_0
t_f
t

Path planning (global):

- Geometric path.
- Issues: obstacle avoidance, shortest path.

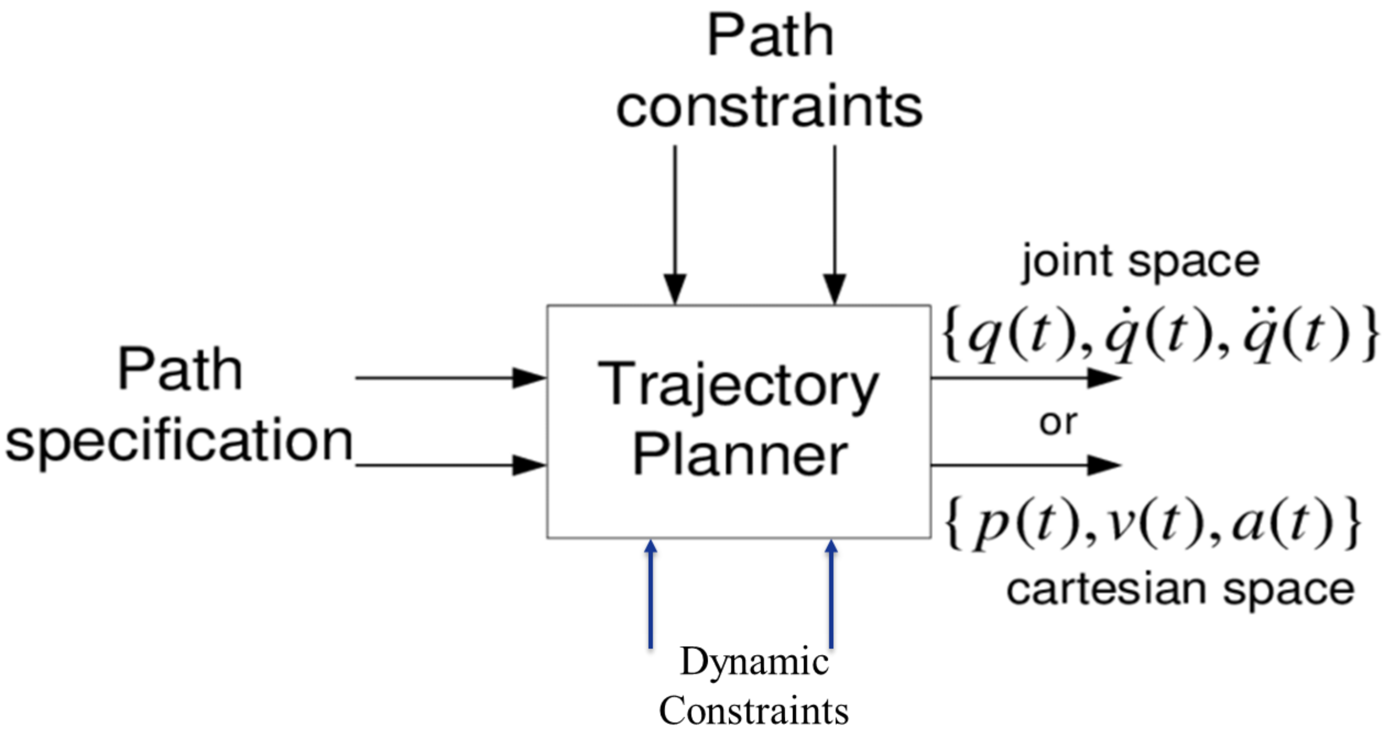
Trajectory generating (local):

- “Interpolate” or “approximate” the desired path by a class of polynomial functions and
- generate a sequence of time-based “control set points” for the control of manipulator from the initial configuration to its destination.



Trajectory planning

Input and output



Path Specification: For generating trajectory points, with the use of polynomial analytical function.

Path Constraints: The intermediate points generated have to be connected to make the path continuous and smooth so that the position and the time derivatives like velocity and acceleration in the path are realistic and traversable.

Dynamic Constraints: Involve the usage of joint torque and force control in the process of path generation and control.

Output will be time sequence of points with location, velocity and acceleration.

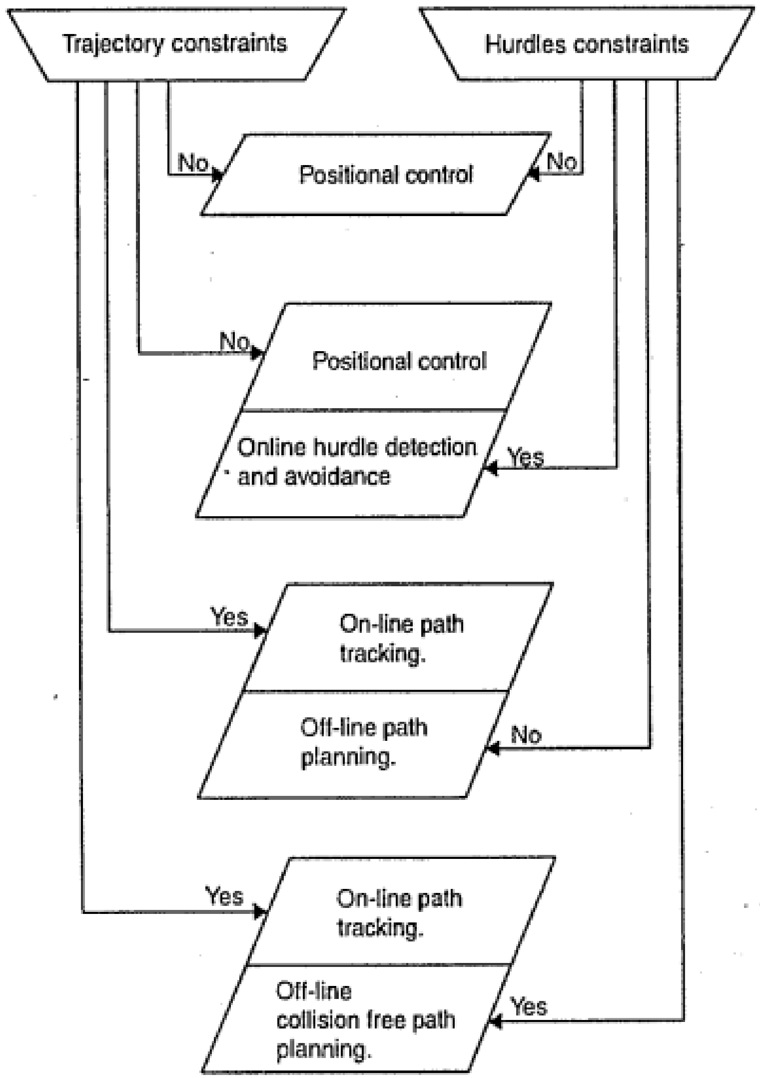
Trajectory planning

Control Modes

i) Trajectory Constraints ii) Obstacles Constraints

- **Mode 1:** When there is no obstacle and there is no constraint on the path to be followed, the positional control is sufficient
- **Mode 2:** When there is certain obstacle and no path constraint, the positional control is to be employed with hurdle detection and avoidance.
- **Mode 3:** When there is no obstacle in the path and tool has to move in certain given path, the off line path planning with on line path tracking has to be carried out.
- **Mode 4:** The presence of a hurdle with a definite constraint on path needs off line collision free path planning coupled with on line path tracking.

Control Modes



Trajectory planning

General consideration in trajectory planning:

- **Frame Basis:** It is advantageous to describe and generate path by specifying robot in terms of tool frame with respect to the base frame.
- **Specification of path:** The specification of a path or trajectory should be task-dependent and independent of the robot.
- **Movements of manipulator:** The manipulator motion is identified and relative to the initial and final position of the tool frame with respect to the base frame.
- **Elapse Time:** The specifying of elapsed time between intermediate points in the description of the path gives temporal attributes to the manipulator motion. (A temporal attribute is a time-related data point in the real world or in a model)
- **Smooth traverse:** To describe the motion the path has to be described by a well defined smooth function which is also continuous with two of its time derivatives. A trajectory should be smooth to avoid wear on joints, motors, and gears.

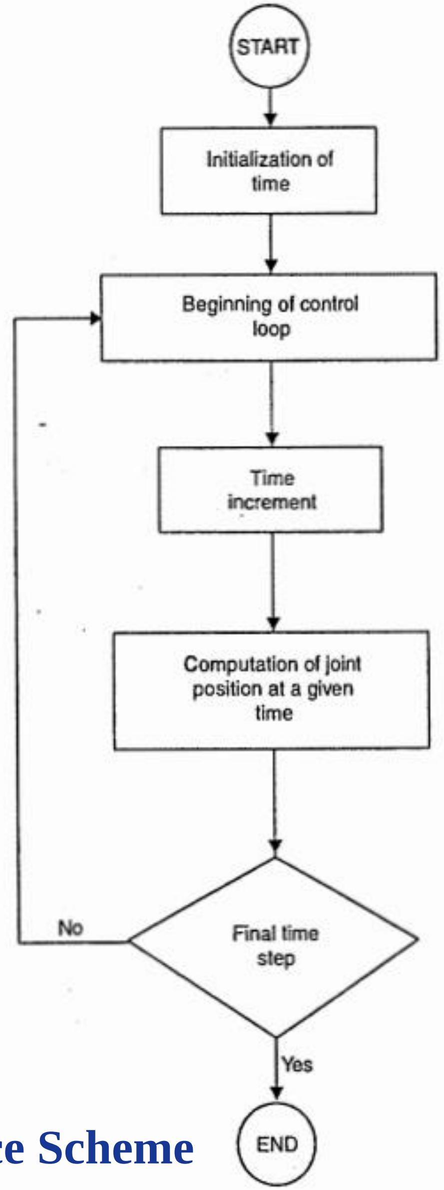
Trajectory planning

Schemes of path generation:

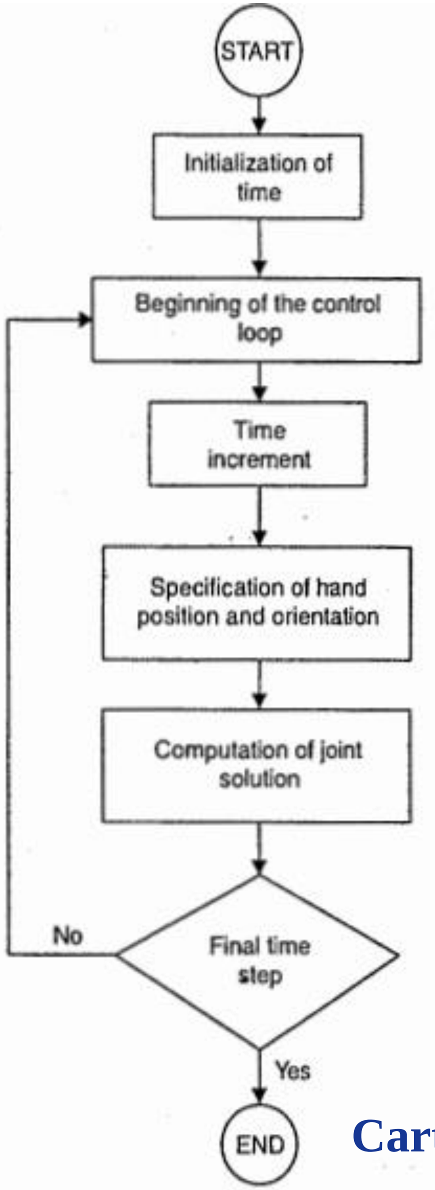
- **Joint Space Scheme:** In this the time history of all joint variables and their first two time derivatives are used for planning the motion of the manipulator.(Forward kinematics)
- **Cartesian Space Scheme:** In this method the manipulators hand position, velocity and acceleration time history are used in planning and using these joint space variables are computed. (Inverse kinematics)

Type of Scheme	*Advantages*	*Disadvantages*
1. Joint space scheme	• Planning of trajectory from control variables in direct terms. • Real time planning of trajectory. • Ease of generating trajectory planning. • Uses low-degree polynomials to interpolate joint hinge points. • Computationally faster. • Dealing of the manipulator dynamic constraints is relatively easy.	• Difficulty in determining the location of the joints, links and hand during motions. • Obstacle avoidance in the path is very difficult. • Less accurate along the cartesian path.
2. Cartesian space scheme	• The method is straight forward. • The assured accuracy along the straight line path. • Easy determination of link and had locations during motion. • Obstacles can be easily avoidable.	• Computationally intensive. • Longer control intervals. • Mapping the hand co-ordinates into joint co-ordinates is not properly defined. • The problem in optimization because of mixed constraint specifications in path and dynamic variables.

Trajectory planning



Joint Space Scheme



Cartesian Space Scheme

Trajectory planning

General consideration of Joint Interpolated trajectory:

- **Hand Motion Direction:** In the object pick up, the motion of the hand has to be directed away from the position of the object, to prevent the crashing of the end effector into the object support structure.
- **Admissible Departure Motion:** The departure motion requires the specification of lift-off point along the normal vector to the surface out from the initial position and if we require the hand to pass through this position, we then have admissible departure motion. If we further specify the time required to reach this position, we could then control the speed at which the object is to be lifted.
- **Set-down motion:** Requirement for this type of motion is same as that for departure motion from the lift-off point, to achieve the correct approach direction and control.
- **Position in trajectory:** In the path to be generated by the hand there are four positions: (1) Initial 2)Lift-off 3)Set-down 4)Final
- **Work space Limitation:** The extrema of all joint trajectories must be within the physical and geometric limits of each joint.

Trajectory planning

General consideration of Joint Interpolated trajectory:

- **Position Constraints:** a) Initial position: Velocity and acceleration are given(normally assumed zero) b) Lift-off-position: Continuous motion for intermediate points c) Set down point: Continuous motion for intermediate points d) Final position: Velocity and acceleration are given (normally assumed zero)
- **Time constraint: a)**Initial and final trajectory segments: time is based on Rate of approach of the hand to and from the surface and Actuator characteristics b)Intermediate points or midtrajectory segment: time is based on maximum velocity and acceleration of the joints.

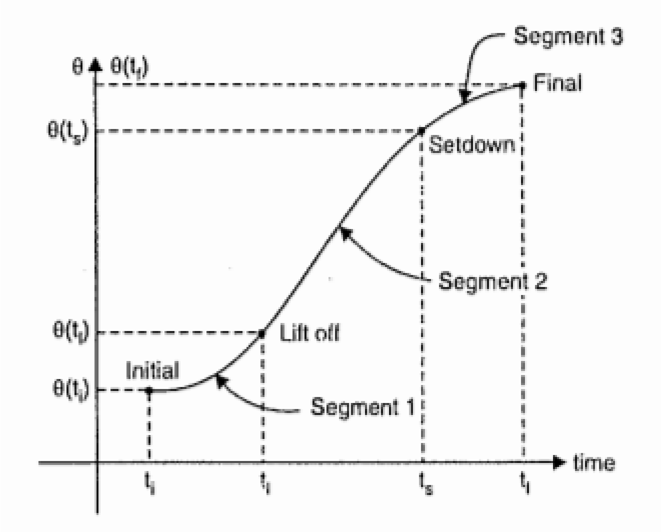


Fig. 6.5. Joint Trajectory Positions.

Trajectory planning

6.5 TRAJECTORY GENERATION PLANNING

Joint Space Trajectory Planning		*Cartesian Trajectory Planning*
Low-order Planning	*Higher-order Planning*	
• By third-order polynomial. $\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$	• 4–3–4 trajectory – Segment 1 : 4^{th} degree polynomial. – Segment 2 : 3^{rd} degree polynomial. – Segment 3 : 4^{th} degree polynomial.	• Segmented differential motion : Smooth straight line transformation between initial and final position by fine segmentation of the path.
• By fifth order polynomial. $\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$.	• 3–5–3 trajectory – Segment 1 : 3^{rd} degree polynomial. – Segment 2 : 5^{th} degree polynomial. – Segment 3 : 3^{rd} degree polynomial. • 5-Cubic trajectory 5 segments of 3^{rd} degree.	• Motion by T2R motion : The transformation between initial and final positions is divided into one translation and two rotations. • Motion by 1T1R motion : The transformation between initial and final position is decomposed into a translation and a rotation about an arbitrary axis.

Trajectory planning

6.7 TRAJECTORY PLANNING WITH 3rd ORDER POLYNOMIAL

To describe the path traced by the hand of the robot a polynomial of the following form is assumed and four set of boundary conditions are imposed to solve the problem to arrive at the joint variables.

- The polynomial :

$$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \quad \text{...(6.6)}$$

- The boundary conditions are imposed on equation (6.6)

at $t = t_i = 0$; $\theta(t) = \theta(t_i) = a_0$.

at $t = t_f$; $\theta(t_f) = a_0 + a_1 t_f + a_2 t_f^2 + a_3 t_f^3$

Time derivatives,

at $t = t_i = 0$; $\dot{\theta}(t_i) = 0, a_1 = 0.$

at $t = t_f$; $\dot{\theta}(t_f) = 0.$

By differentiating equation (6.6) twice w.r.t. 't'

$$\dot{\theta}(t) = \frac{d\theta}{dt} = a_1 t 2a_2 t + 3a_3 t^2$$

and

$$\ddot{\theta}(t) = \frac{d^2\theta}{dt^2} = 2a_2 + 6a_3 t$$

By knowing starting and finish joint angle and intermediate time the joint angles for various positions can be obtained. The respective joint velocities and the accelerations can also be computed.

Example 6.1. *One of the joint of a articulated robot has to traverse from an initial angle of 20° to a final angle of 84° in 4 seconds. Using a third degree polynomial calculate the joint angle at 1, 2 and 3 second.*

Sol. The given third degree polynomial is

$$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

The time derivatives given by

$$\dot{\theta}(t) = a_1 + 2a_2 t + 3a_3 t^2$$

$$\ddot{\theta}(t) = 2a_2 + 6a_3 t$$

At $t = t_i = 0$, $\theta(t_i) = a_0 = 20$

$$\dot{\theta}(t_i) = a_1 = 0 \text{ (starting velocity zero)}$$

At $t = 4 = t_f$, $\theta(t_f) = a_0 + a_1(4) + a_2(4)^2 + a_3(4)^3 = 84$

$$\dot{\theta}(t_f) = a_1 + 2a_2(4) + 3a_3(4)^2$$

The simultaneous equations are

$$16a_2 + 64a_3 = 64$$

$$8a_2 + 48a_3 = 0$$

The solution of the simultaneous equation yields

$$a_2 = -12 \text{ and } a_3 = -2$$

The polynomial is $\theta(t) = 20 - 12t^2 - 2t^3$

$$\dot{\theta}(t) = -24t - 6t^2$$

At the given times

t	1	2	3
θ	6°	–44°	–142°

Trajectory planning

- Path Profile
- Velocity Profile
- Acceleration Profile